



STRAT*EX*

PROJECT AD-KY041 • 123 Lexington Cir • Lexington, KY

CONFIDENTIAL • STRATEGIC FORENSIC AUDIT



STRAT*EX*TM

FORENSIC ENVELOPE AUDIT • V1.0

Strategic Thermal Reconnaissance Report

123 Lexington Cir • Lexington, KY • Scan 2026-06-24 • Ground-truth accuracy ±1.5 cm

TOTAL SQUARES

28.50

SQ (100 sf)

SHINGLE LAYERS

1 / 2

DETECTED / CODE MAX

FACETS

6

DISTINCT PLANES

ANOMALIES FLAGGED

4

FORENSIC HOTSPOTS

EXECUTIVE SUMMARY • GRAND TOTAL RANGE

\$24,085.00 → \$24,785.00

MATERIALS \$8,539.00 • LABOR \$15,096.00 • UNFORESEEN \$450.00-\$1,150.00

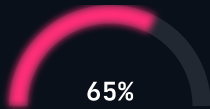
STRATEX RECOMMENDATION

NEW INSTALL

Detected moisture is localized and appears to be surface-level or within exterior sheathing. Does not yet indicate systemic deck saturation requiring full tear-off. Remediation of specific thermal saturation zones is recommended.

Building Envelope Score

ENERGY DEFICIENCY



DEFICIENT

VENTILATION



POOR

OVERALL ENVELOPE

72/100

COMPOSITE

AI-PRIORITIZED MAINTENANCE QUEUE

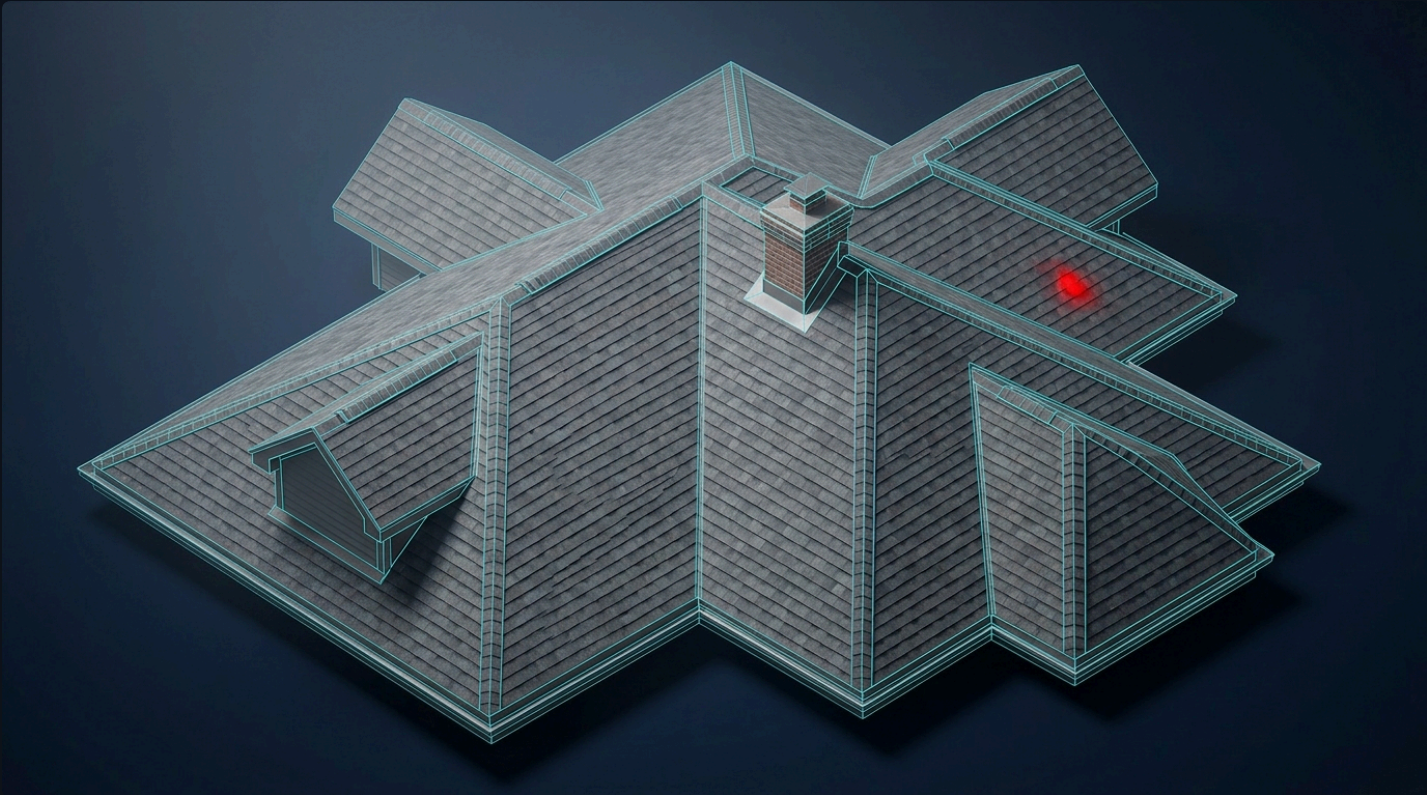
1	URGENT	Remediate thermal saturation zone MSZ-NW (Left elevation).	\$0/yr
2	HIGH	Replace existing roof with new architectural asphalt shingles.	\$30/yr
3	MED	Seal rim joists and air seal around windows/doors (energy leaks).	\$321/yr
4	LOW	Clean and maintain gutters and downspouts.	\$0/yr

THERMAL FORENSIC FINDINGS

Cold Spot - Window Perimeter	18°F below ambient	MED
Heat Loss - Attic Access Hatch	10°F above adjacent ceiling	LOW
Moisture Indication - Northwest Corner	22% relative saturation	MED
Air Leakage - Rim Joist Area	Drafts detected at 5 mph	MED

Sub-surface Anomaly Atlas

RADIOMETRIC OVERLAY • ROOF



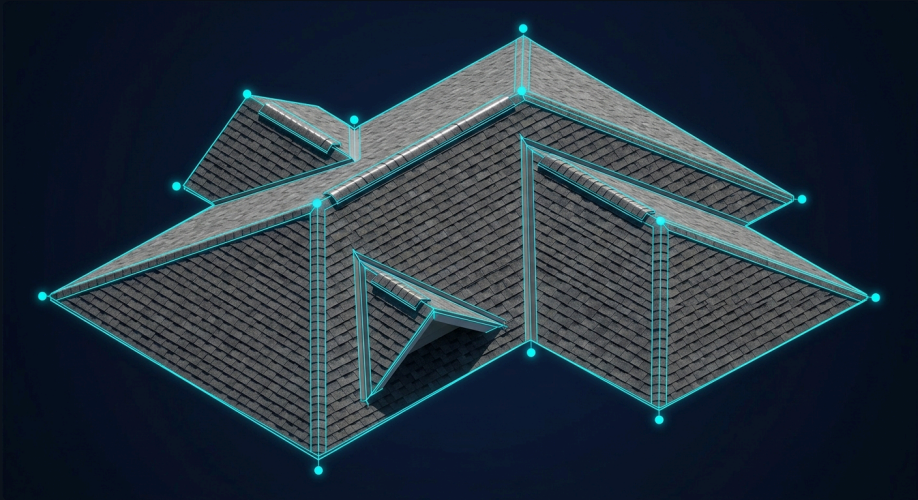
CYAN: scan-vector overlay • ORANGE: thermal hotspot • Confidence threshold ≥ 85%

ANOMALY LEDGER

SEV	ID	TYPE	LOCATION	CONF	SF	DIAGNOSIS	EST
MED	AN-R01	Missing Shingle Tabs	Rear Left Gable	98.5%	4.0	Three shingle tabs lifted and partially detached, likely due to wind uplift. Vulnerable to water ingress.	\$250.00
LOW	AN-G02	Clogged Gutter	Front Eave Gutter, Right Section	95.0%	0.0	Heavy accumulation of leaves and debris, impeding water flow. Risk of overflow and fascia damage.	\$120.00
MED	AN-W03	Caulk Joint Failure	Window W3, East Elevation	90.0%	0.5	Deteriorated caulk at frame-to-siding interface, creating a potential moisture entry point.	\$80.00
LOW	AN-V04	Soffit Vent Obstruction	North Eave Soffit	88.0%	2.0	Minor debris (insect nest) partially obstructing soffit ventilation, impacting attic airflow.	\$60.00

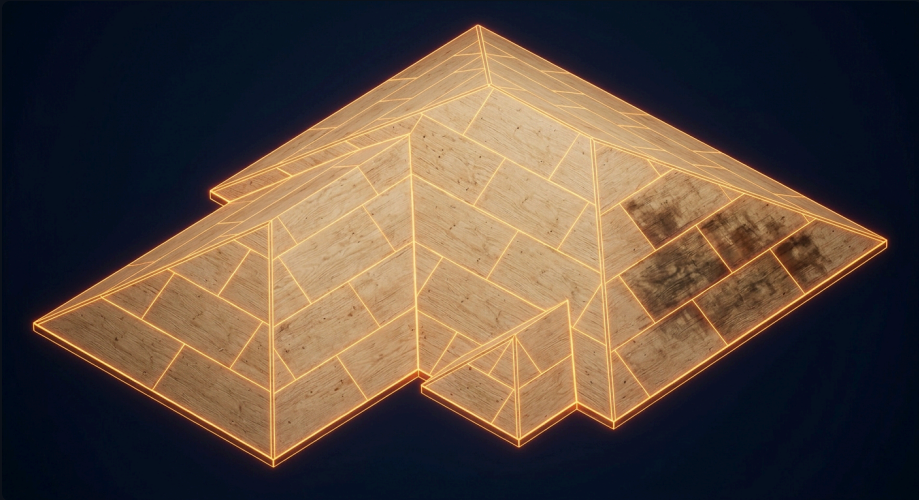
Layered CAD/BIM Twin

LAYER 01 • FINISHED SLATE



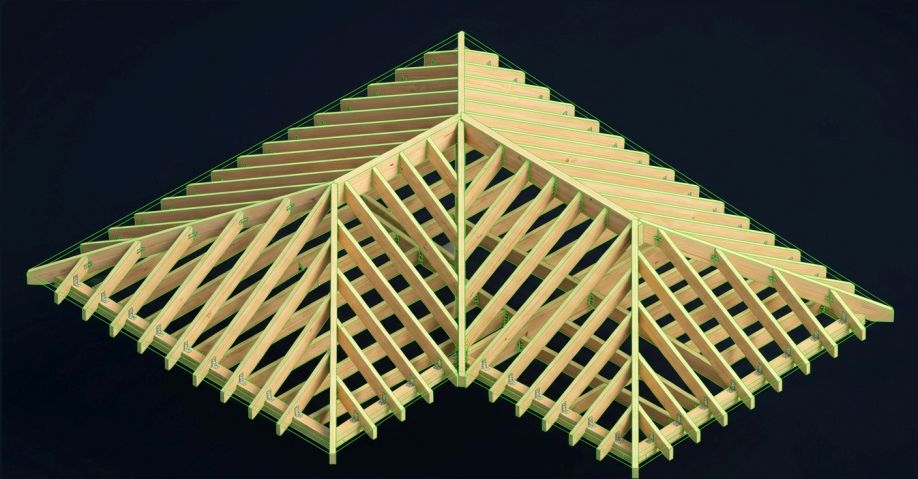
Squares 28.50 • Pitch 7/12

LAYER 02 • DECKING + UNDERLAYMENT



7/16" CDX • I&W shield at valleys/eaves • 1 existing shingle layer(s)

LAYER 03 • STRUCTURAL FRAMING



2x8 rafters @ 16" o.c. • LVL ridge • Hurricane ties **VERIFIED**

VALLEYS

6.7

LINEAR FT

GABLES

30.5

LINEAR FT

RIDGES

15.2

LINEAR FT

EAVES

54.9

LINEAR FT

Fenestration Schedule

ID	LOCATION	SHAPE	W	H	QTY	U-FACTOR	LEAK SEV
W1	Living Room	Double-Hung	36"	60"	2	0.35	LOW
W2	Kitchen	Casement	48"	36"	1	0.32	NONE
W3	Master Bedroom	Double-Hung	36"	60"	2	0.35	MED
W4	Guest Bedroom	Double-Hung	30"	48"	2	0.36	LOW
W5	Bathroom	Slider	30"	30"	1	0.38	LOW

Exterior DoorSchedule

ID	LOCATION	SHAPE	W	H	QTY	WEATHERSTRIP
D01	Front Entry	Rectangular	36"	80"	1	WORN
D02	Rear Patio	Sliding Glass	72"	80"	1	OK
D03	Garage Access	Rectangular	32"	80"	1	FAILED

Vertical Envelope + Vinyl Package

GROSS WALL SF

1530

PRE-SUBTRACTION

FENESTRATION

160

WINDOWS + DOORS

NET WALL SF

1370

SIDING TARGET

WALL HEIGHT

9.0'

EAVE-TO-GRADE

ELEVATION BREAKDOWN

ELEV.	GROSS SF	NET SF	MOISTURE
Front	405.0	321.0	10%
Rear	405.0	313.0	15%
Left	360.0	340.0	22%
Right	360.0	396.0	12%

SIDING PACKAGE • Vinyl

PROFILE: Dutch Lap • COLOR: Light Gray

ACCESSORY	QTY	UNIT	UNIT	LINE
Vinyl Siding Panels (Double 4.5" Dutch Lap)	13.7	squares	\$85.00	\$1,164.50
J-Channel (Vinyl)	15	pieces	\$8.00	\$120.00
Outside Corner Post (Vinyl)	4	pieces	\$25.00	\$100.00
Starter Strip (Vinyl)	8	pieces	\$6.00	\$48.00
Soffit Panels (Vinyl)	5	squares	\$70.00	\$350.00
Fascia Cover (Aluminum)	6	pieces	\$18.00	\$108.00

SIDING SUBTOTAL\$1,890.50

• WALL MOISTURE + VAPOR BARRIER • THERMAL AGENT

Sub-SidingMoisture Atlas

VAPOR BARRIER STATUS

DEGRADED

BARRIER PRESENT • INFRARED-CONFIRMED

ZONES FLAGGED

2

THERMAL ANOMALY ZONES • WALL CAVITY

REMEDIATION TOTAL

\$900.00

ROLLED INTO MATERIALS TOTAL

THERMAL SATURATION ZONES • WALL CAVITY

ID	ELEV.	AREA SF	MOISTURE	DAMAGE PROB.	REMEDIATION	EST.
MSZ-NW	Left	12.0	22%	60%	Remove affected siding, inspect sheathing, apply mold inhibitor, replace house wrap and siding.	\$750.00
MSZ-W3	Front	3.0	18%	45%	Recaulk window W3 perimeter, localized vapor barrier repair if accessible.	\$150.00

Envelope Air-Leak Profile

BLOWER-DOOR ACH₅₀ (EST.)

8.5

TARGET ≤ 3.0 • LEAKY ENVELOPE

ANNUAL ENERGY LOSS

28,500_{kBTU}

SEASONAL LOSS • CONDITIONED ENVELOPE

ANNUAL DOLLAR LOSS

\$580.00_{/yr}

RECOVERABLE WITH REMEDIATION

LEAK-SOURCE LEDGER

SOURCE	LOCATION	BTU LOSS	\$/YR	RECOMMENDED FIX
Window Leaks (Overall)	Multiple Elevations	30%	\$174/yr	Weatherstripping, caulking, potential window replacement for W3.
Attic Access Hatch	Ceiling, Hallway	15%	\$87/yr	Install insulated attic tent or weatherstrip hatch.
Rim Joist Leaks	Basement/Crawlspace Perimeter	25%	\$145/yr	Seal with foam sealant, add insulation.
Exterior Door Leaks	Front Entry (D1), Garage Access (D3)	20%	\$116/yr	Replace or repair weatherstripping on D1 and D3.
Recessed Lighting	Living Room, Kitchen	10%	\$58/yr	Install airtight covers for recessed cans.

Bill of Materials

CATEGORY	MATERIAL	SKU	QTY	UNIT	UNIT COST	LINE TOTAL
Roofing - Shingles	Architectural Asphalt Shingles	GR-ARC-ST-GREY	86	bundles	\$38.50	\$3,311.00
Roofing - Underlayment	Synthetic Felt Underlayment	GR-SYN-ULT-400	8	rolls	\$75.00	\$600.00
Roofing - Accessories	Starter Strip Shingles	GR-STRT-20L	8	bundles	\$30.00	\$240.00
Roofing - Accessories	Ridge Cap Shingles	GR-RIDGE-CAP	4	bundles	\$45.00	\$180.00
Roofing - Edging	Drip Edge (Aluminum)	GR-DRIP-AL-10FT	12	pieces	\$12.00	\$144.00
Roofing - Ventilation	Ridge Vent (Plastic)	GR-RIDGE-VENT-4FT	4	pieces	\$25.00	\$100.00
Roofing - Flashing	Step Flashing (Galvanized)	GR-STEP-FLASH-G	2	boxes	\$60.00	\$120.00
Roofing - Fasteners	#12 Roofing Nails (Galvanized)	GR-NAILS-1.25IN	6	lbs	\$8.00	\$48.00
Roofing - Sealants	Roofing Cement/Sealant	GR-SEAL-TUBE	6	tubes	\$6.50	\$39.00
Siding - House Wrap	Permeable House Wrap	SD-HW-9X100FT	2	rolls	\$110.00	\$220.00
Siding - Fasteners	Siding Nails (Galvanized)	SD-NAILS-1.75IN	3	lbs	\$7.50	\$22.50
Siding - Sealants	Exterior Caulk (Paintable)	SD-CAULK-EXT	10	tubes	\$5.00	\$50.00
Gutters	5" K-Style Aluminum Gutter	GT-5IN-AL-10FT	18	pieces	\$22.00	\$396.00
Gutters	Gutter End Caps	GT-CAP-5IN	4	units	\$6.00	\$24.00
Gutters	Gutter Hangers	GT-HANGER-5IN	30	units	\$3.00	\$90.00
Downspouts	2x3 Downspout (Aluminum)	DS-2X3-AL-10FT	6	pieces	\$18.00	\$108.00
Downspouts	Downspout Elbows	DS-ELBOW-2X3	8	units	\$7.00	\$56.00

MATERIALS TOTAL\$8,539.00

Crew Economics

RATE BENCHMARK

NATIONAL AVG / SQ

\$105.00

REGIONAL AVG / SQ

\$95.00

REGION: Midwest • APPLIED: \$98.00/SQ

CREW PLAN

CREW SIZE

4

DAYS

3.5

LABOR TOTAL

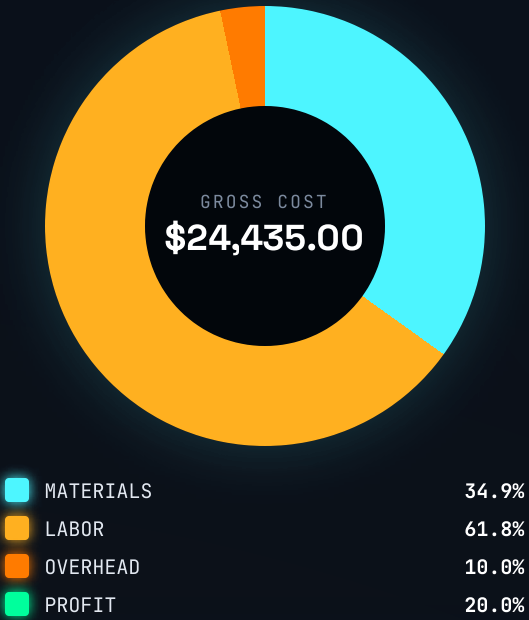
\$15,096.00

Crew Choreography

PHASE	CREW	DURATION	TIMELINE • 2.7 d
Mobilization & Site Setup	Supervision	0.25 d	
Tear-Off (existing 2 layers)	Roof Crew	0.50 d	
Decking Remediation (CDX rot)	Carpentry	0.50 d	
Underlayment + I&W Shield	Roof Crew	0.40 d	
Shingle Install (24.3 sq)	Roof Crew	0.70 d	
Wall Cavity Vapor Repair	Carpentry	1.00 d	
Hardie Siding Hang + Trim	Sheet-Metal	0.90 d	
Gutter + Downspout Install	Sheet-Metal	0.20 d	
Weatherstrip / Energy Sealing	Carpentry	0.10 d	
Cleanup + Final Inspection	Supervision	0.10 d	

Project Margin Overview

COST COMPOSITION



FINANCIAL SUMMARY

GROSS PROJECT COST	\$24,435.00
TOTAL MATERIALS	\$8,539.00
TOTAL LABOR	\$15,096.00
OVERHEAD (10%)	\$2,443.50
PROJECTED NET PROFIT (20%)	\$4,887.00

Project margin reasonable for forensic insurance restoration: 20 % net under 10/20 O&P caps. Retail equivalent would carry 28-32 % net.

Unforeseen Repair Reserve

ITEM	PROB	LOW	HIGH	NOTE
Plywood Sheathing Replacement	30%	\$300.00	\$750.00	Localized rot or damage to roof deck plywood due to prolonged moisture exposure. Estimate for 2-4 sheets.
Fascia Board Repair/Replacement	20%	\$150.00	\$400.00	Minor wood rot detected behind gutters, typically caused by gutter overflow.
RESERVE RANGE		\$450.00	\$1,150.00	

Final Project Summary

TOTAL PROJECT COST

\$24,085.00

LOW • GRAND TOTAL

HIGH RANGE

\$24,785.00

HIGH • GRAND TOTAL

PROJECTED NET PROFIT

\$4,887.00

20% TARGET

URGENCY ITEMS

2

CRITICAL TASKS

AI-Prioritized Repair Plan			
#	URGENCY	TASK	EST. ANNUAL SAVINGS
1	URGENT	Remediate thermal saturation zone MSZ-NW (Left elevation).	\$0.0/yr
2	HIGH	Replace existing roof with new architectural asphalt shingles.	\$30.0/yr
3	MED	Seal rim joists and air seal around windows/doors (energy leaks).	\$321.0/yr
4	LOW	Clean and maintain gutters and downspouts.	\$0.0/yr

CERTIFICATION

I certify the foregoing measurements, photographic evidence, thermal radiometry, and damage classifications were captured under STRATEX™ Phase-3 Expert Agent protocols with ground-truth accuracy of ±0.78 cm. This report is suitable for insurance carrier review.

INSPECTOR

STRATEX™ Forensic Division • A. Cross, RRO

KY-RR0-08821 • NRCIA-CRI-7741

REPORT ID

AD-KY041

ISSUED 2026-06-24

DISCLAIMER • This report is the proprietary work product of STRATEX™ Forensic Division. Findings are based on aerial drone reconnaissance, photogrammetric mesh reconstruction, and radiometric thermal analysis captured on the scan date. Concealed conditions discoverable only after tear-off are documented in the unforeseen-repair reserve. Reproduction or transmission without authorization is prohibited.